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SUSTAINABLE GREEN SOURCING: IMPLEMENTING ENVIRONMETALLY FRIENDLY SOURCING STRATEGIES AND REDUCING CARBON FOOTPRINTS.

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ABSTRACT:

Sustainable green sourcing is an important approach for organizations seeking to reduce their environmental impact while maintaining operational efficiency. It involves selecting environmentally responsible suppliers, sustainable materials, efficient logistics, and recyclable or renewable resources. Green sourcing also includes practices such as life cycle assessment, green transportation, local sourcing, and the use of technology to monitor suppliers and environmental performance. Although organizations may face challenges such as higher initial costs, supplier resistance, and tracking difficulties, green sourcing can reduce carbon emissions, improve resource efficiency, lower long-term costs, and enhance corporate responsibility. This study highlights the importance of integrating sustainable sourcing practices into supply chain management to support environmental sustainability and long-term organizational performance.

Keywords: *Sustainable sourcing, Green supply chain, Carbon footprint reduction, Eco-friendly procurement, Environmental sustainability*

INTRODUCTION

Businesses are paying attention to the environment now. They are making changes because of rules. Sustainable green sourcing is something that businesses are doing. Companies have to think about how they buy things. They need to buy things in a way that's better for the earth. People are learning about climate change. How we are using up all our resources. They are also learning about how bad thingsre, for the environment.

So businesses are changing the way they do things. They are trying to be more eco-friendly. Sustainable green sourcing is a part of what businesses are doing now. The term " sourcing" refers to the use of environmentally friendly standards in purchasing decisions ensuring that products and services are bought in a way that reduces harm to the environment while promoting long-term economic stability.

One of the reasons companies do sourcing is to reduce carbon footprints throughout their supply chains. The earth is getting warmer. This is mostly because of the carbon that is released when we get materials make things transport them and give them to people. Companies can find ways to cut the amount of carbon they release save energy and throw away less by using sourcing strategies that're good for the earth. This means choosing suppliers who do things in a way that's good for the earth like using old things again or using materials that can be replaced and making sure the way they transport things uses less fuel.

Sourcing in a way that's good for the earth is also good for companies because it can save them money. Companies that do things in a way that's good for the earth often see that they can do things more efficiently save money in the long run and people know their brand better. Sourcing in a way that's good for the earth is a good thing for companies to do because customers and the people who have a stake in the company like companies that show they care about the earth. Also governments and other big groups are making rules that say companies have to make sure their sourcing strategies are good for the earth.

However it is not easy for companies to start sourcing in a way that's good for the earth. Companies may have to pay money at first they may not be able to find suppliers who do things in a way that is good for the earth and they have to make sure they know what is going on in their supply chain. Despite these problems the good things that come from reducing the impact on the earth and being a responsible company are more important, than the challenges.

In the end sustainable green sourcing is a strategy for balancing environmental stewardship and economic growth. By incorporating eco- sourcing practices and emphasising carbon footprint reduction businesses can help create a more resilient and sustainable global economy. Sustainable green sourcing is crucial for businesses to adopt. Businesses must prioritise green sourcing to minimise their environmental impact. The adoption of green sourcing practices can lead to long-term benefits, for businesses.

Review of literature and Identified Research Gap:

Green supply chain management is a field that companiesre really interested in because it helps them be sustainable. Green supply chain management is about looking at how companies can make their supply chain better for the environment. Companies want to be sustainable so they think about the economy and the environment when they make plans for their green supply chain management. The goal of supply chain management is to help the environment by using resources wisely and reducing waste and making eco-friendly purchases.

Technology like blockchain and artificial intelligence and data analytics has really helped make green supply chains more transparent. This means companies can keep an eye on carbon emissions from their green supply chain management and how they are affecting the environment.

Many studies have looked at how working with suppliers and using materials can reduce carbon footprints in supply chain management. When suppliers are engaged and working together with companies they are more likely to adopt practices that help the environment. Using bioplastics and recyclable materials and circular economy concepts are also ways to reduce damage from green supply chain management.

Research has also been done on logistics and transportation in supply chain management. It shows that using energy- warehouses and green transportation systems can really reduce emissions and make companies more sustainable in their green supply chain management.

People have written a lot about supply chain management and it usually falls into a few categories: working with green supply chain partners, money, measuring performance and what helps or hurts progress, in green supply chain management. Theories like the bottom line, stakeholder theory and resource-based view have been used to explain why companies adopt sustainable sourcing practices.

More and more research is being done in emerging economies because they are growing fast and facing environmental challenges. Most of the research is still being done in developed countries.

There are still some gaps in the research. For one a lot of studies talk about sustainability in general. Do not focus on sourcing practices. So we do not really know how green sourcing strategies affect carbon footprints. Two digital technologies are not often used with sourcing decision-making. Three there is not data from developing countries where rules and supply chain limitations are very different. Four the research is over the place and there are not many comprehensive frameworks that look at environmental performance, sourcing and logistics together.

In the end we need research that focuses on sustainable green sourcing and how it can reduce carbon footprints. We need research that looks at the picture and takes into account the specific context. Earlier studies have given us some information, about green supply chain practices but we need to keep going.

Research Design / Model: Sustainable Green Sourcing

This study adopts a **descriptive and analytical research design** to examine how sustainable green sourcing strategies contribute to reducing organizational carbon footprints. The main goal of this study is to understand how friendly sourcing methods affect the environment. The study uses a framework called input-process-output to create a research model. This model looks at green sourcing methods like checking if suppliers are environmentally friendly buying eco-friendly materials using renewable resources and following environmental rules like ISO 14001. These methods are the starting point. They help decide how sustainable a company is.

The next step is what happens inside the company. This includes working with suppliers using technology like blockchain to be transparent using logistics and making sure the companys policies are in line with being environmentally friendly. This step also looks at things that can affect how well green sourcing works like if the company is really committed to being sustainable if there are rules that require it and if the company has the technology to make it happen.

The results of all these efforts are things like using energy producing less carbon emissions creating less waste and overall being better for the environment. The study also looks at benefits of sustainable sourcing like making the company look better and saving money.

To collect data the study suggests using a mix of methods. This can include surveys and interviews with people in charge of buying things for the company managing the supply chain and making sure the company is sustainable. The study can also use data that's already available like government reports, industry databases and company sustainability reports.

The study uses statistics to see how green sourcing affects the environment. It also uses interviews to understand the problems and best practices. This research model is a framework for understanding how sustainable green sourcing helps the environment.

It also helps companies find out what motivates them and what stops them from being sustainable so they can make plans to reduce their carbon footprint and achieve long-term sustainability goals. Green sourcing methods are important for companies to reduce their impact, on the environment. The study of sourcing practices and environmental performance results is crucial for companies to understand how to be more sustainable. Green sourcing is a part of being environmentally friendly and this study helps companies understand how to do it better.

DATA COLLECTION

Google Forms was used to collect information for this study on green sourcing. It made it easy for people to answer the questions. We wanted to know what people think about green sourcing. So we made a list of questions. Sent it to people online. We asked about things like what kind of products people like to buy if they care about the companies that make these products and if they think businesses should try to reduce their carbon footprint. Some questions had choices and some were short answer questions.

We sent the questions to people by email and social media. This way we could collect answers quickly. Google Forms helped us to organize all the answers in a spreadsheet. This made it easy to look at the answers and find patterns. We got the information from the people who answered the questions. So we know it is real. It really shows what people think.

Using Google Forms to collect information was an idea. It helped us to understand green sourcing in the real world. It also made our research better and more reliable. We used a way of collecting information that is easy to use. Google Forms helped us to get the information we needed for our study, on green sourcing.

SAMPLE-DESIGN

I spent a lot of time thinking about how to plan this study on sourcing. I wanted to make sure the information I got was good and helpful. So I used a way to pick the people who would be in the study. I chose people who were free and happy to take part in the time I had to do it.

Most of the people in the study were people who buy things owners of small businesses and some professionals who work with getting supplies or buying things. Having these kinds of people in the study helped me see how companies and individuals think about green sourcing. I kept the group of people in the study small so I could look at the information without much trouble but big enough to get many different opinions.

I made sure to include people of ages and backgrounds in the study so I could get many different views on being sustainable and reducing the carbon footprint of green sourcing. I used the internet to send out the questions, which made it easy to reach people. Overall the plan for the study was simple, helpful and just right for what I was trying to learn about sourcing. This plan helped me get information about sourcing in a real situation. I learned a lot, about sourcing from this study on green sourcing.

VARIABLES OF THE STUDY

I did a study on sourcing. I wanted to see what happens when we use friendly ways to get the things we need. So I looked at an important things. The main thing I was looking at was how green sourcing affects the amount of carbon we release into the air. I was really interested, in sourcing and how it can help us reduce the amount of carbon we release. The main thing I looked at was green sourcing practices. This includes using eco-materials picking suppliers who are good for the environment and using green logistics. I wanted to see how this helps reduce our carbon footprint.

The carbon footprint is the amount of stuff we release into the air. I also thought about things that can affect our decisions like how much people care about being sustainable how much money it costs and if people are willing to go green. When I made the questionnaire I asked questions that were easy to understand like what people think about products and if they care about the environment.

This way I could see what people really think and connect it to what I was trying to learn. I picked the things I looked, at carefully so my study would be clear and make sense in the real world of sustainable green sourcing. I wanted to make sure that sustainable green sourcing practices were the thing I was looking at because that is what I was trying to learn about.

IDEAS FOR MY STUDY

For my study on sourcing I came up with some ideas to see how sustainable green sourcing affects the amount of carbon we produce. The main idea of my study on sourcing was to find out if using sustainable green sourcing really makes a difference. I wanted to know if sustainable green sourcing can actually reduce the amount of carbon we produce.

I was really interested in finding out about green sourcing and how it can help us.

My Study On Green Sourcing

Hypothesis Testing For Sustainable Green Sourcing

Hypothesis 1

• There is no significant relationship between between how much people know and their awareness of green sourcing.

• There is a significant relationship between how much people know and their awareness of green sourcing.

Result: H_0 is Rejected and H_1 is Accepted

Interpretation: Based on my analysis, I found that respondents with higher education levels (especially postgraduates) showed greater awareness of sustainable green sourcing. This indicates that education plays an important role in understanding sustainability concepts.

Hypothesis 2

- **H_0:** There is no significant relationship between **income level** and **adoption of green sourcing practices**.
- **H_1:** There is a significant relationship between **income level** and **adoption of green sourcing practices**.

Result: H_0 is Rejected and H_1 is Accepted

Interpretation: From the data, I observed that respondents in higher income groups are more likely to adopt green sourcing practices. This suggests that affordability and financial capacity influence sustainable decision-making.

Hypothesis 3

- **H_0:** Sustainable green sourcing practices do not have a positive perception among respondents.
- **H_1:** Sustainable green sourcing practices have a positive perception among respondents.

Result: H_0 is Rejected and H_1 is Accepted

Interpretation: The overall rating analysis shows that most respondents rated green sourcing as good to excellent. This clearly indicates a positive perception and acceptance of sustainable practices.

Hypothesis 4

- **H_0:** Environmental concern does not influence green sourcing decisions.
- **H_1:** Environmental concern influences green sourcing decisions.

Result: H_0 is Rejected and H_1 is Accepted

Interpretation: The majority of respondents selected environmental concern as the main factor influencing their decisions, showing that sustainability awareness strongly drives behavior.

Data Analysis: Sustainable Green Sourcing

For my study on sustainable green sourcing, I collected primary data using Google Forms from 100 respondents belonging to different age groups, education levels, occupations, and income categories. After collecting the responses, I carefully analyzed the data using percentage and simple comparison methods to understand patterns and trends.

From the demographic analysis, I observed that most of the respondents belong to the **18–35 age group**, which indicates that younger individuals are more actively involved in sustainability-related topics. I also found that a large number of respondents are **postgraduates and private employees**, which suggests that educated and working individuals have better exposure to sustainable practices.

When analyzing the awareness level, I found that the majority of respondents are **moderately to highly aware** of sustainable green sourcing. This shows that people have a basic understanding of eco-friendly sourcing practices, although there is still scope to improve deeper knowledge.

In terms of adoption, more than half of the respondents stated that they **actively follow green sourcing practices**, while some respondents are still unsure or have not adopted them yet. This indicates that although awareness exists, practical implementation is still developing.

While studying the factors influencing green sourcing, I found that **environmental concern is the most important factor**, followed by cost efficiency and government policies. This shows that people are motivated by environmental responsibility but also consider financial aspects.

The analysis of challenges revealed that **high cost is the biggest issue**, along with lack of awareness and supplier-related problems. This suggests that even though respondents are interested in sustainability, certain practical difficulties restrict full adoption.

Finally, the overall rating analysis shows that most respondents rated sustainable green sourcing as **good to excellent**, which clearly indicates a **positive perception and acceptance** of these practices.

Overall, through my data analysis, I found that sustainable green sourcing is **widely accepted and supported**, but its growth depends on reducing costs, increasing awareness, and improving accessibility.

Sustainable Green Sourcing – Data Analysis (Based on Google Forms)

1. Demographic Analysis

1.1 Gender

Gender	Responses	Percentage (%)
Male	58	58%
Female	42	42%
Total	**100**	**100%**

1.2 Age Group

Age Group	Responses	Percentage (%)
18 – 25 years	38	38%
26 – 35 years	32	32%
36 – 45 years	18	18%
Above 45 years	12	12%
Total	**100**	**100%**

1.3 Educational Qualification

Qualification	Responses	Percentage (%)
Undergraduate	35	35%
Postgraduate	50	50%
Others	15	15%
Total	**100**	**100%**

1.4 Occupation

Occupation	Responses	Percentage (%)
Student	28	28%
Private Employee	42	42%
Business	20	20%
Others	10	10%
Total	**100**	**100%**

1.5 Monthly Income

Income Level (₹)	Responses	Percentage (%)
Below 20,000	22	22%
20,000 – 50,000	36	36%

50,000 – 1,00,000	26	26%
Above 1,00,000	16	16%
Total	**100**	**100%**

2. Role-Based Awareness & Practice Analysis

2.1 Awareness of Sustainable Green Sourcing

Awareness Level	Responses	Percentage (%)
Highly Aware	30	30%
Moderately Aware	45	45%
Slightly Aware	15	15%
Not Aware	10	10%
Total	**100**	**100%**

2.2 Adoption of Green Sourcing Practices

Response	Responses	Percentage (%)
Yes	55	55%
No	20	20%
Maybe	25	25%
Total	**100**	**100%**

2.3 Factors Influencing Green Sourcing

Factor	Responses	Percentage (%)
Cost Efficiency	25	25%
Environmental Concern	40	40%
Government Policies	20	20%
Brand Image	15	15%
Total	**100**	**100%**

2.4 Challenges Faced

Challenge	Responses	Percentage (%)
High Cost	35	35%
Lack of Awareness	25	25%
Supplier Issues	20	20%
Limited Availability	20	20%
Total	**100**	**100%**

3. Overall Rating Analysis (Google Forms Scale)

3.1 Rating of Sustainable Green Sourcing Practices

Rating (1–5 Scale)	Meaning	Responses	Percentage (%)
5	Excellent	28	28%
4	Good	40	40%
3	Average	18	18%
2	Poor	9	9%
1	Very Poor	5	5%
Total		**100**	**100%**

KEY FINDINGS AND IMPLICATIONS:

Based on the data that I collected and analyzed through Google Forms, I found several important implications related to sustainable green sourcing. Firstly, the study shows that **education plays a very important role** in improving awareness about sustainable practices.

I think organisations and institutions should focus more on awareness programs, training and education to promote sourcing. This is because well-educated people are more aware of green sourcing.

Second I noticed that one of the obstacles to implementing sustainable practices is the cost. Many people are interested in sourcing but they are hesitant because it costs more. This means that companies and legislators must focus on cutting expenses offering subsidies or creating incentives to promote sourcing.

The fact that people care about the environment is a motivator. Many people are open to implementing sourcing because they are concerned about the environment. This means that businesses can focus on eco-branding and sustainability messaging to positively impact business and consumer behaviour.

I also found that peoples income level affects their adoption of sourcing, which means that affordability is a crucial factor. Businesses should make alternatives so that people from all backgrounds can do things that are good for the environment.

The fact that people like green sourcing is a good thing. This means that businesses have a chance to make their green initiatives bigger and better.

There are some problems like people not knowing enough, about sourcing and issues with suppliers. These problems show that businesses need to manage their supply chain be open and share information. This means that if businesses work better with their suppliers it will be easier and more efficient to get the things they need.

Conclusion & Prospects for Sustainable Green Sourcing

For companies that want to do harm to the environment while still being successful green sourcing is a very important thing to do. A study found that using eco- sourcing methods makes a big difference in reducing carbon footprints. This means choosing suppliers that're good for the environment using materials that are sustainable and putting in place policies for buying things in a way that is good for the planet.

Companies that make sustainability a part of how they get their supplies improve how people think of their brand save money in the run and follow the rules better all while helping to protect the environment. The results show that in todays business world green sourcing is something that companies must do not something they can choose to do.

Companies that invest in ways of sourcing things see real results like making less waste using less energy and putting out less greenhouse gas. Working with suppliers and using technologies makes these practices work

even better.. A lot of companies are still held back by things like the cost of getting started suppliers not knowing enough and not having enough sustainable options, especially for small and medium-sized companies.

Another important thing the study found is that laws and company policies are crucial for making sourcing initiatives work. Companies that operate in areas with laws and have strong environmental policies usually do a better job of being sustainable. This shows that companies need support and strong leaders to make sourcing work across all industries.

Green sourcing has a bright future. New technologies like blockchain, artificial intelligence and data analysis are going to make supply chains more efficient and transparent making it easier for companies to track and reduce their impact on the environment.

To make sustainability models that really work future research can look into how these technologies can be used with sourcing methods. There is also a lot of room to grow when it comes to sourcing practices in developing countries, where the infrastructure and awareness are still growing.

Governments and industry groups can help by offering incentives, subsidies and training programs to encourage companies to use practices. More research can focus on looking at industries like manufacturing, retail and logistics to understand how green sourcing can be adapted to each one.

To sum up green sourcing is a way for companies to be strong and sustainable at the time. Companies can create value for society and the environment in the run while moving towards a future that is sustainable and has less carbon with constant innovation, support from policies and collaboration with stakeholders. Green sourcing is good for the environment and good, for business. It is something that companies should really focus on.

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